
Zev-7

Nixie Clock

Assembly Manual

May 2026

This manual describes the assembly, testing, and operation of the Zev-7 Nixie Clock from the bare PCB and components.

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Table of Contents

1. Introduction 1-1

2. Safety Warnings 2-1

3. Required Tools and Equipment..... 3-1

4. Assembly Sequence 4-1

5. Initial Power-On and Testing 5-1

6. Troubleshooting6-1

Appendix A — Bill of MaterialsA-1

Appendix B — Schematics.....B-1

1. Introduction

This manual covers the assembly of the Zev-7 Nixie Clock from bare PCB and components. The Zev-7 uses seven Nixie tubes driven by three HV5530 high-voltage shift registers, an ESP32 microcontroller for WiFi and NTP timekeeping, a high-voltage boost converter, and addressable RGB LED backlighting.

The circuit is documented across four schematic sheets reproduced in Appendix B:

- Power Supply (ps.dxf) — 12V input, 5V regulator, 170V HV supply
- Processor (cpu.dxf) — ESP32, level shifter, Schmitt trigger, sensors, buttons
- Display (nixie.dxf) — HV5530 shift chain, Nixie tubes, neon indicator lamps
- RGB LEDs (led.dxf) — APA106 tube backlights, APA102 under-lights

Components should be soldered in the order presented in Section 4, progressing from low-profile parts to tall components, and from power to signal to display.

2. Safety Warnings

WARNING: This circuit generates approximately 170V DC for the Nixie tube anodes. This voltage is potentially lethal. Never touch any part of the high-voltage circuit while the clock is powered on. Always disconnect power and wait at least 30 seconds for capacitors to discharge before working on the board.

WARNING: Nixie tubes are glass and fragile. Handle tubes by the base only. Do not apply excessive force when inserting tubes into sockets.

CAUTION: Observe correct polarity on all electrolytic capacitors (C1, C2, C3), diodes and ICs. Reversed polarity can cause immediate component damage.

CAUTION: The LM2576T-5.0 and Tayloredge 1363 module may become hot during operation. Allow adequate ventilation and do not touch metal tabs during operation.

NOTICE: If for any reason the above conditions and the copyright notice at the end of this manual cannot be met, please return the kit or assembled clock in original condition within 30 days of purchase for a full refund, less shipping.

3. Required Tools and Equipment

Soldering

- Temperature-controlled soldering iron (recommended 350°C / 660°F)
- Fine solder (0.5–0.8 mm rosin-core, 60/40 Sn/Pb)
- Solder flux (liquid or paste)
- Solder wick or desoldering pump
- Tweezers (fine-tip, for SMD components)
- Magnifying visor or bench magnifier

Test Equipment

- Digital multimeter (must be capable of measuring DC voltage up to 200V)
- 12V DC power supply (minimum 1A, center-positive)
- Oscilloscope (optional, useful for debugging data signals)

Hand Tools

- Wire strippers and flush cutters
- Anti-static wrist strap

4. Assembly Sequence

Assemble in the order below, progressing from lowest-profile components to tallest, and from power to signal to display. Test each section before proceeding to the next.

4.1 Step 1: Power Supply

Build and test the power supply first to ensure clean voltage rails before installing sensitive ICs.

Install Components

1. F1 (1 A fuse)
2. C1 (330 μ F 16V) — Observe polarity.
3. LM2576T-5.0 — Tab is oriented toward the outside of the board.
4. D2 (1N5819) — Cathode band as marked on PCB.
5. L1 (100 μ H)
6. C2 (330 μ F 16V) — Observe polarity.
7. R1 (10k), R22 (39k)
8. J1 (DC barrel jack)

Test the 5V Rail

Connect a 12V supply to J1. Measure the output with a multimeter:

- VCC / VDD rail should read $5.0\text{V} \pm 0.1\text{V}$
- If the voltage is incorrect, check D2 orientation and L1 connections.

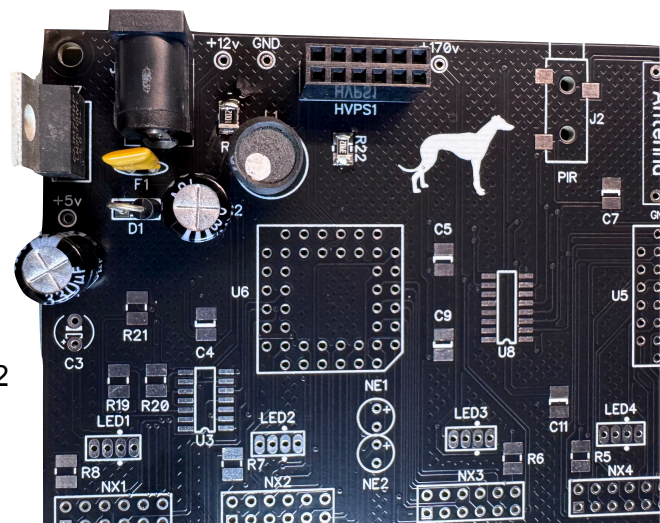
Disconnect power before proceeding.

Install HV Supply

9. Solder the 2x6 Header and insert HVPS1 (Tayloredge 1363) — Temporarily connect a wire between pin 10 and +5V to enable the power supply.

Test

Apply 12V power. Check the 170V, 12V, and 5V test points with a multimeter. The HV supply will read approximately 172V. Remove the temporary wire.



4.2 Step 2: Processor and Support ICs

SMD Components First

Solder all SMD parts before through-hole components.

1. C3 (100 μ F), C4-C11 (100 nF ceramic) — Decoupling and bulk capacitors. Solder these first as they are the lowest-profile components.
2. R14 (39k), R15 (2.7k), R16 (2.7k), R17 (10k), R18 (10k)
3. U3 (74HCT14D, SOIC-14) — Verify pin 1 alignment.
4. U8 (CD4504BM, SOIC-16) — Verify pin 1 alignment.

Through-Hole Components

5. U1 (ESP32 PICO) — Do not solder the module to the board! Cut the single row header strip to fit, and solder in place.
6. Q23 (TEPT4400) — Lens faces outward toward ambient light. Shorter lead toward the flat side of the silkscreen.
7. D3 (1N5819) - Note orientation.

4.3 Step 3: Display Drivers (HV5530)

1. U4-U6 (HV5530, PLCC-44) — Carefully orient the three sockets. Orient the socket so the flat corner is on the front-right of the board (to match the silkscreen markings). Install the three HV5530 chips in the same orientation. They will click into place.

WARNING: Do not get this wrong. It is extremely difficult to remove these sockets once soldered, and requires specialized equipment.

4.4 Step 4: Nixie Tubes and Neon Lamps

WARNING: The anode resistors connect to the 170V HV rail. Double-check all connections before applying power. A short on the HV rail can damage components instantly.

Anode Resistors

1. R2–R8 (15k $\times$ 7) — Anode resistors.

Neon Lamp Resistors

2. R9–R13 (270k $\times$ 5) — Neon lamp resistors (on the underside of the PCB)

Nixie Tube Sockets, Boards, and Tubes

3. Install tube sockets — Solder sockets for NX1–NX7. Do not insert tubes until the HV rail has been verified.
4. Different tube types are available. Each tube type requires a specific tube board. Carefully solder each tube to a tube board, noting the side on which to install the tube. First install the male connector to the tube board, then the tube on the other side.
5. Install neon lamps — NE1–NE5.

Tube Position Reference

Tube	Position	Displays	Anode Resistor	Driven By
NX1	Leftmost pair	Hours tens	R8	U6
NX2		Hours ones	R7	U6
NX3	Middle pair	Minutes tens	R6	U6
NX4		Minutes ones	R5	U5
NX5	Rightmost pair	Seconds tens	R4	U5
NX6		Seconds ones	R3	U5
NX7		Tenths of second	R2	U4

Neon Lamp Reference

Lamp	Position	Signal	Resistor	Function
NE1	Between NX1/NX2	TLNeon	R9	Top-left colon
NE2	Between NX1/NX2	BLNeon	R10	Bottom-left colon
NE3	Between NX3/NX4	TMNeon	R11	Top-middle colon
NE4	Between NX3/NX4	BMNeon	R12	Bottom-middle colon
NE5	Near NX7	BRNeon	R13	Right indicator

4.5 Step 5: RGB LEDs

Chain 1: Tube Backlights (APA106)

1. R19 (150Ω)
2. LED1–LED7 (APA106) — The longer pins go toward to dot on the silkscreen. Before soldering, temporarily install a tube board and push the LED up into the hole in the board as far as it will go.

Chain 2: Under-Lights (APA102)

3. R20 (150Ω), R21 (150Ω)
4. LED8–LED11 (APA102) — Note the orientation, these are on the bottom of the PCB

CAUTION: APA106 and APA102 LEDs have different pinouts. Carefully follow the silkscreen when installing.

4.6 Step 6: Sensors

BME280 Temperature/Pressure/Humidity Sensor

1. J3 (4-pin header) — BME280 I²C sensor connector.

The BME280 is auto-detected at I²C addresses 0x76 and 0x77. The sensor is optional — the clock operates normally without it.

PIR Motion Sensor

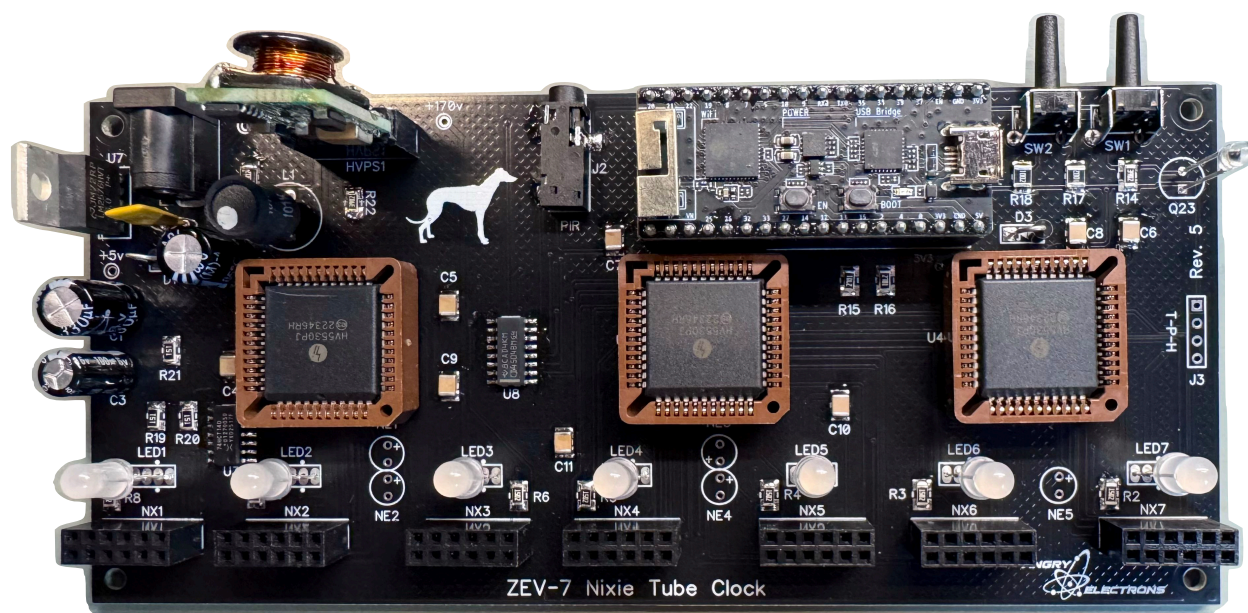
2. J2 (PIR connector)

Light Sensor

3. Q23 (TEPT4400) — Already installed in Step 2.

4.7 Step 7: Buttons and Connectors

1. SW1 (Pushbutton)
2. SW2 (Pushbutton)



The Completed Board - Without Colons Installed

5. Initial Power-On and Testing

| WARNING: Do not insert Nixie tubes until the HV rail has been verified.

Pre-Power Checks

Before applying power for the first time:

- Visually inspect all solder joints under magnification
- Check for solder bridges, especially on SOIC and PLCC packages
- Verify all electrolytic capacitor polarities (C1, C2, C3)
- Verify diode orientations (D2, D3)
- Confirm no shorts between power rails: measure resistance between VCC and GND, HV and GND

Power-On Sequence

1. Connect 12V supply to J1. Current draw should be under 100 mA with no tubes installed.
2. Verify voltages:
 - VCC / VDD: $5.0V \pm 0.1V$
 - VEE (ESP32 3.3V rail): $3.3V \pm 0.1V$
 - HV rail: 170V (no adjustment needed)
3. The ESP32 should create a WiFi access point named "Zev-7" upon first boot.
4. Connect from a cell phone or PC to the clock WiFi access point and then open a browser window at 192.168.4.1. At a minimum select your WiFi network and enter the network password. You can change settings here on the various pages and press Save. Press Back, and then Start, on the home page to initiate the clock.
5. After WiFi connects, the display shows the startup sequence, and starts running.

Testing Subsystems

LEDs: After WiFi connection and time sync, the RGB backlights should activate with the default "Random Multi" effect. If LEDs do not light, check R19, R20, R21, U3, and data line continuity.

Buttons: Press SW1 — should trigger a button override. Hold SW2 for 5 seconds — should restart in AP mode.

Sensors: If BME280 is connected, enable it in Setup. Temperature, humidity, and pressure appear in the info display cycle. Triple-click SW2 to verify the light sensor (raw reading on NX1–NX2).

6. Troubleshooting

Symptom	Check	Solution
No power (nothing lights)	F1 fuse, J1 connection	Replace fuse. Check barrel jack center-positive.
5V rail incorrect	LM2576 connections, D2 orientation, L1	Verify LM2576 pinout. Check D2 Schottky cathode direction. Confirm L1 value.
ESP32 not booting	VEE (3.3V), solder joints on module	Reflow ESP32 module pins. Check for shorts on IO0 (must be high for normal boot).
LEDs not lighting	R19, R20, R21, U3 (74HCT14D), VCC to LEDs	Verify 5V at LED VCC pins. Check R19 / R20 / R21 continuity. Verify U3 pin 14 = 5V.
LED chain partially works	Failed LED in chain	Data cascades: a dead LED breaks the chain. Replace the first non-working LED.
No display (tubes dark)	HVENABLE, HV rail, R2–R8	Verify HVENABLE is high when display should be on. Check HV voltage. Verify anode resistors.
Wrong digits displayed	DATA / CLK / LE wiring, U8 level shifter	Verify U8 connections. Check shift chain order (U6→U5→U4). Confirm CLK / LE / DATA signals with scope.
Digits dim or flickering	HV voltage, BL (blanking) signal	Verify HV rail at 170V. Verify BL PWM signal from IO13 through U8.
Neons not working	R9–R13, HV5530 outputs for neons	Verify 270k resistors. Check that neon outputs toggle (use scope on HV5530 output pins).
No WiFi AP on first boot	ESP32 flash, antenna	Reflash firmware via USB. Ensure no metal near ESP32 antenna area.
BME280 not detected	J3 wiring, I ² C pull-ups R15 / R16	Verify SDA / SCL at J3. Check R15 / R16 to 3.3V. Try alternate I ² C address module.

PIR not responding	J2 wiring, C7, IO37	Verify signal at IO37. Check PIR module power (3.3V). Confirm C7 decoupling.
Light sensor not reading	Q23 orientation, R14	TEPT4400 must face ambient light. Verify R14 (39k) and signal at IO38.

Appendix A — Bill of Materials

Component values reflect the latest schematic revisions in Appendix B. Reference designators match the silkscreen on the PCB.

Integrated Circuits

Ref	Part	Package	Qty	Description
U1	ESP32 PICO KIT V3	Module	1	ESP32-PICO-D4 WiFi microcontroller
U3	74HCT14D	SOIC-14	1	Hex inverting Schmitt trigger
U4	HV5530	PLCC-44	1	HV shift register (NX7 + BR neon)
U5	HV5530	PLCC-44	1	HV shift register (NX4, NX5, NX6 + TM/BM neons)
U6	HV5530	PLCC-44	1	HV shift register (NX1, NX2, NX3 + TL/BL neons)
U8	CD4504BM	SOIC-16	1	CMOS hex voltage level shifter
—	LM2576T-5.0	TO-263-5	1	5V 3A switching regulator
HVPS1	Tayloredge 1363	Module	1	HV boost converter (~170V)

Resistors

Ref	Value	Qty	Description
R1	10k	1	HV voltage divider
R2–R8	15k	7	Nixie tube anode current limiters
R9–R13	270k	5	Neon lamp current limiters
R14	39k	1	Light sensor (TEPT4400) load resistor
R15, R16	2.7k	2	I ² C pull-ups (SDA, SCL)
R17	10k	1	HVENABLE pull-down
R18	10k	1	Button pull-up (SW2)
R19	150Ω	1	APA106 data line series resistor
R20	150Ω	1	APA102 data line series resistor
R21	150Ω	1	APA102 clock line series resistor
R22	39k	1	39k resistor

Capacitors

Ref	Value	Qty	Description
C1	330 μF 16V	1	Input filter (12V rail), electrolytic
C2	330 μF 16V	1	Output filter (5V rail), electrolytic
C3	100 μF	1	ESP32 supply bulk capacitor
C4–C11	100 nF	8	Decoupling capacitors (ceramic)

Diodes

Ref	Part	Qty	Description
D2	1N5819	1	Schottky catch diode (LM2576)
D3	Protection diode (1N5819)	1	VCC protection (near ESP32)

Inductors and Fuses

Ref	Value	Qty	Description
L1	100 μ H	1	LM2576 buck converter output inductor
F1	1 A fuse	1	Input power fuse

Nixie Tubes and Neon Lamps

Ref	Part	Qty	Description
NX1–NX7	Nixie tube	7	Numeric display tubes (digits 0–9)
	12 Pin Male Socket	7	Mounts to PC Board
	12 Pin Female Socket	7	Mounts to the bottom of the tube boards
NE1–NE5	Neon indicator lamp	5	Colon and status indicators

LEDs

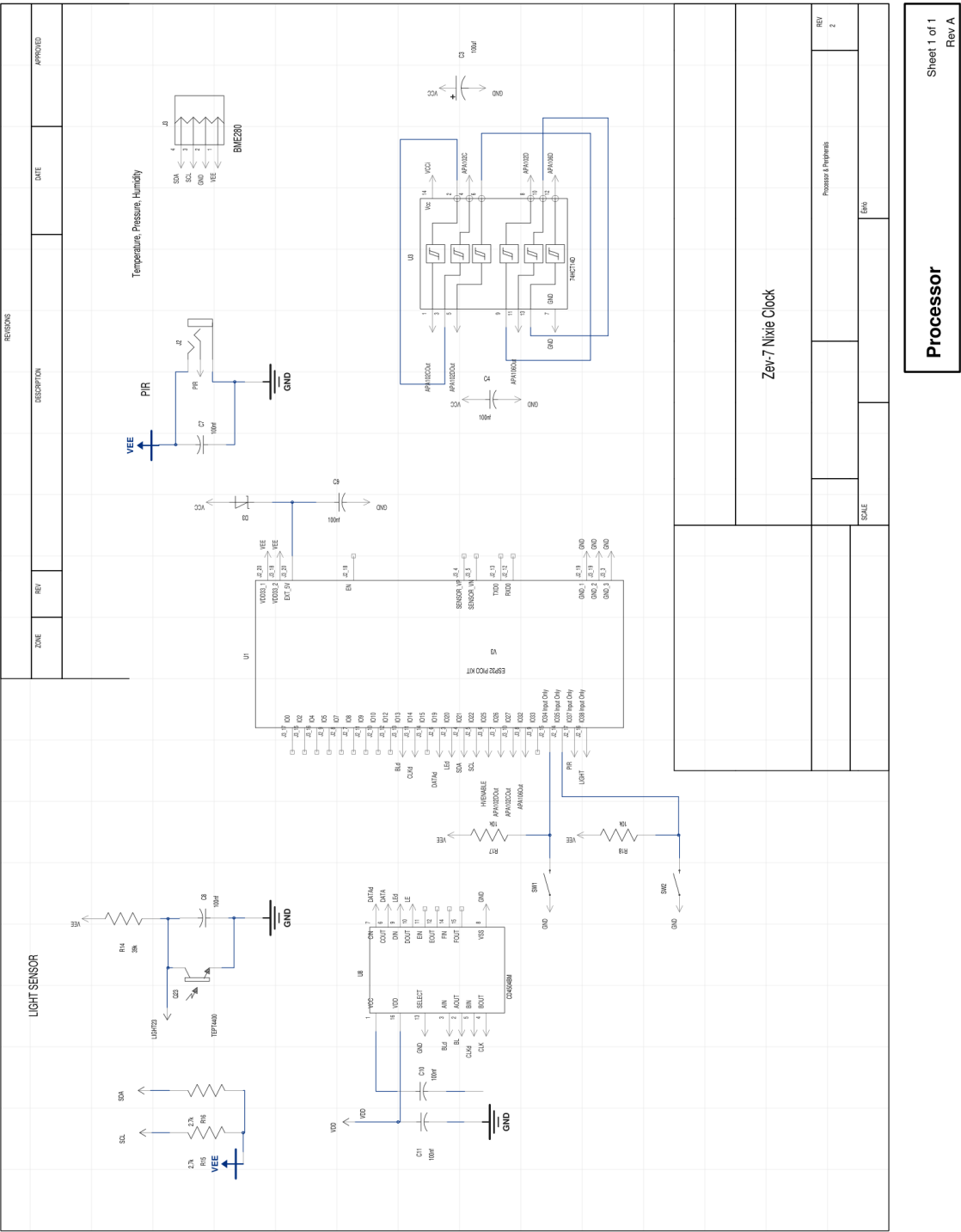
Ref	Part	Qty	Description
LED1–LED7	APA106 RGB LED	7	Tube backlight (single-wire, IO5)
LED8–LED11	APA102 RGB LED	4	Under-light (SPI, IO32 / IO27)

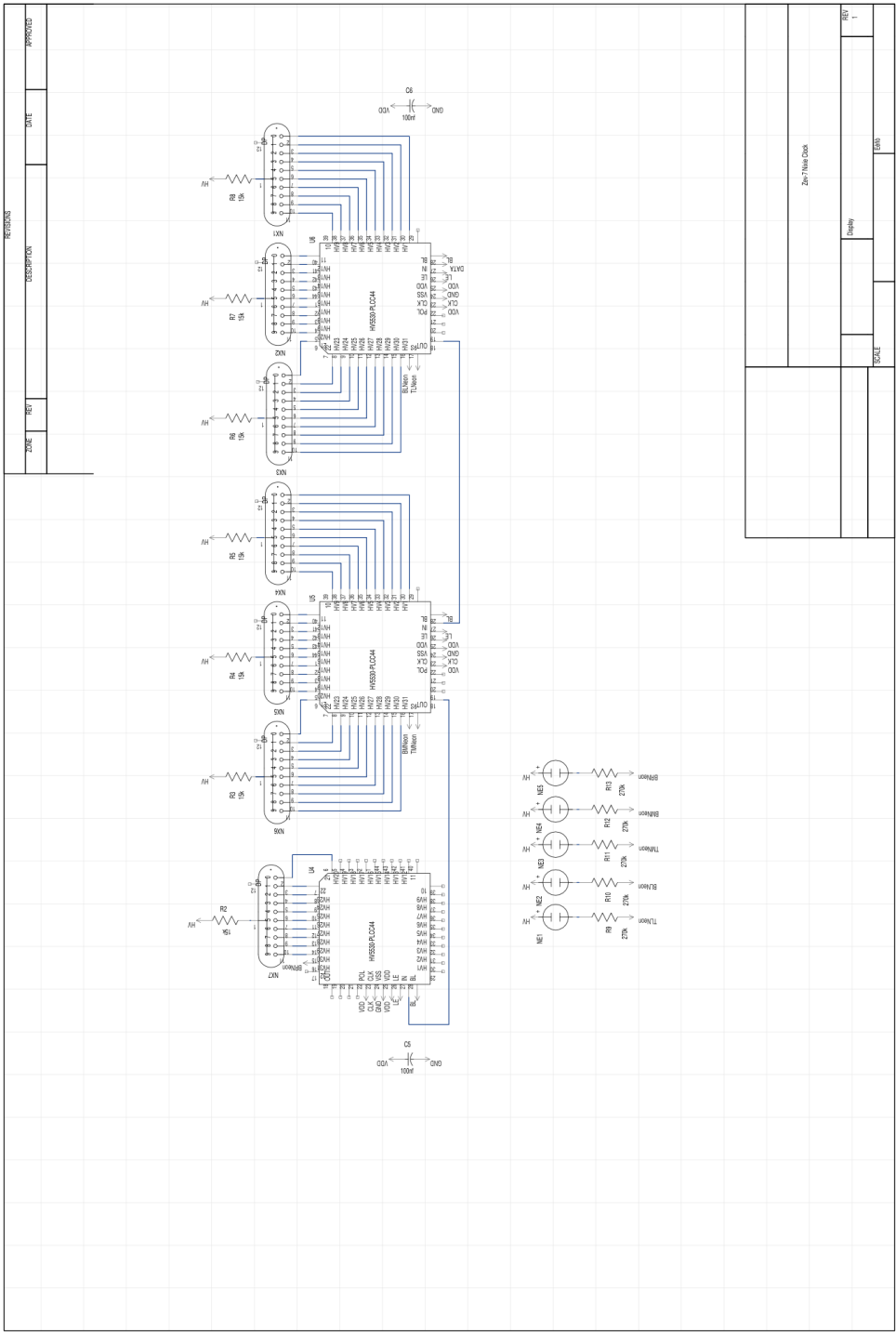
Connectors, Switches, and Sensors

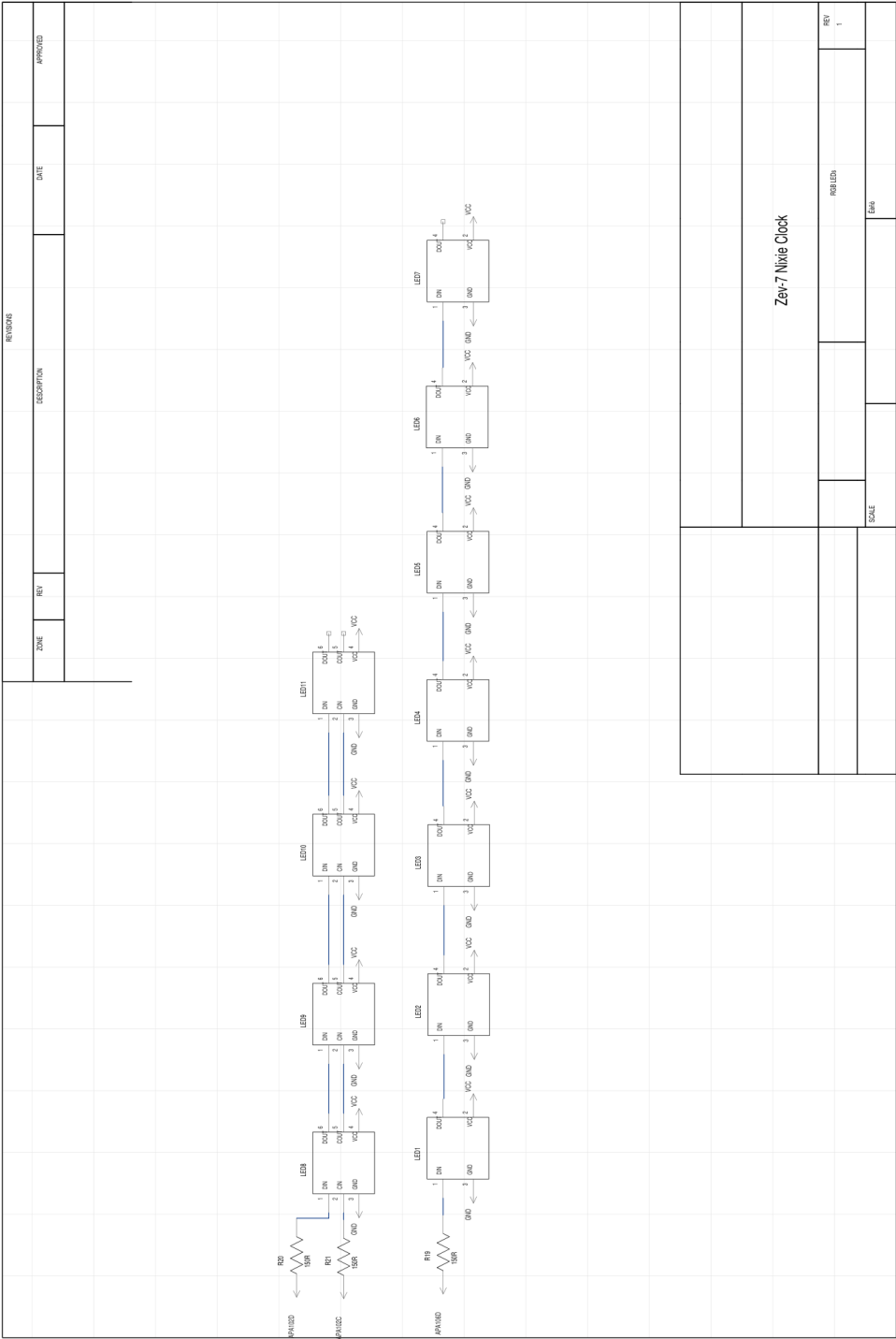
Ref	Part	Qty	Description
J1	DC barrel jack	1	12V power input
J2	PIR connector	1	PIR motion sensor header
J3	4-pin header	1	BME280 I ² C sensor connector
SW1	Pushbutton	1	Button 1 (override, IP, OTA)
SW2	Pushbutton	1	Button 2 (calibration, AP mode)
Q23	TEPT4400	1	Ambient light phototransistor

Appendix B — Schematics

Each of the four schematic sheets is reproduced in landscape orientation on its own page on the pages that follow. Sheets are presented in the assembly order: Power Supply, Processor, Display, RGB LEDs.







RGB LEDs

Sheet 1 of 1

Rev A

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The Nixie tubes used in this clock contain a small amount of mercury and neon gas under low pressure. The high-voltage circuitry inside generates approximately 170 volts and should not be touched while the clock is powered on.

Zev-7 Nixie Clock — Designed by Mitchell Feig